

SPRING CORE AND MAVEN

EXERCISE 1

Objective

To configure a basic Spring Core application using XML configuration, define beans for the service and repository layers, and verify the configuration by loading the Spring application context.

BookRepository.java

```
package com.library.repository;

public class BookRepository {
    public void displayRepository() {
        System.out.println("Book Repository is working.");
    }
}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void displayService() {
        System.out.println("Book Service is working.");
        bookRepository.displayRepository();
    }
}
```

applicationContext.xml

```
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://www.springframework.org/schema/beans
        http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```

<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>

<bean id="bookService"
      class="com.library.service.BookService">
  <property name="bookRepository" ref="bookRepository"/>
</bean>

</beans>

```

MainApp.java

```

package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class MainApp {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.displayService();
    }
}

```

Output

The application executed successfully and displayed the following output:

```

Jun 29, 2026 02:15:18 PM org.springframework.context.support.AbstractApplicationContext prepareRefresh
INFO: Refreshing org.springframework.context.support.ClassPathXmlApplicationContext@4b1210ee

Jun 29, 2026 02:15:18 PM org.springframework.beans.factory.xml.XmlBeanDefinitionReader loadBeanDefinitions
INFO: Loading XML bean definitions from class path resource [applicationContext.xml]

Book Service is working.
Book Repository is working.

Jun 29, 2026 02:15:18 PM org.springframework.context.support.AbstractApplicationContext doClose
INFO: Closing org.springframework.context.support.ClassPathXmlApplicationContext@4b1210ee

<terminated> MainApp [Java Application]

```